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Article

Fostering Creative Instructional Design: Unpacking the Role of Metacognitive Scaffolding in an AI Pedagogical Agent for Pre-Service Preschool Teachers—A Moderated Mediation Model

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Abstract

Despite advances in AI pedagogical agents, research on developing creative instructional design remains limited, and how they affect pre-service teachers' creative thinking—especially with metacognitive scaffolding—is unclear. Based on metacognitive and creativity investment theories, this study examines how such scaffolding in an AI agent fosters creative instructional design among pre-service preschool teachers, and whether critical thinking mediates this relationship moderated by AI dependency. A quasi-experimental design was used with 120 pre-service preschool teachers, who were assigned to the experimental group or the control group. Data on metacognitive awareness, AI dependency, critical thinking, and creative thinking were gathered through valid measuring instruments, and innovative curricula were evaluated by experts. The results show that the experimental group achieved much better scores on creative instructional design ideas after the test than the control group. The moderated mediation analysis revealed a critical thinking-mediated pathway that was moderated by AI dependency. In conclusion, AI pedagogical agents with metacognitive scaffolding (MAS-based) improved critical thinking and promoted deeper, more independent creative thinking, thus improving creative instructional design, through a pathway that is moderated by the degree of AI dependency. This study offers valuable theoretical and practical insights to cultivate creative teaching skills.

Keywords: metacognitive scaffolding; creative instructional design; AI pedagogical agent; pre-service preschool teachers; creative thinking; AI dependency

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1. Introduction

Creativity is regarded as one of the most important skills needed to face the challenges of the 21st century (Lee, 2023; Yao et al., 2023). Pre-service preschool teachers need to have the ability to create innovative curricula and inspire children's imagination so they can be effective teachers and cultivate students' creativity (Li, 2023; Yildirim & Yilmaz, 2023). Meanwhile, AI is drastically changing education. AI-driven instructional agents—especially pedagogical agents that give personalized guidance and instant feedback—are becoming popular and open up new possibilities for creativity training in teacher education (Baglivo et al., 2023; D. Chen et al., 2024). By mimicking human teaching behaviors, these systems not only offer academic support, but can also transform the role of teachers, who will no longer be content to simply impart knowledge, but will cultivate a higher level of creativity (Boye & Agyei, 2023).

However, a significant gap persists between theory and practice. Firstly, traditional teacher training has long emphasized rote learning and standardized assessment, resulting in the systematic neglect of fostering creativity (Su et al., 2024). Secondly, despite the rapid advancement of AI technology, AI-based tutoring systems are primarily utilized for conveying subject knowledge and acquiring procedural skills, failing to adequately support higher-order cognitive abilities such as creative instructional design. Furthermore, research on the effective integration of AI within teacher training remains extremely limited at present (Marrone et al., 2022). A particularly noteworthy issue is that metacognitive strategies, widely recognized as the foundation for creative thinking, are not sufficiently incorporated into existing AI tutoring systems (Jiang et al., 2023; McNeely-White & Cleary, 2023). Moreover, AI adoption entails risks: over-reliance can standardize thought and reduce students' enthusiasm for independent investigation and exploration (Yu, 2023; Ahmed et al., 2024). This "AI dependence" might moderate the relationship between MAS and creative results, but there is not much strong evidence on how different kinds of dependence affect it.

Based on the identified research gaps, this study aims to achieve the following objectives: ① to develop and empirically examine the effectiveness of an AI pedagogical agent integrated with metacognitive scaffolding; ② to unveil the underlying mechanisms and delineate the boundary conditions through which metacognitive scaffolding within the AI pedagogical agent influences creative instructional design. To this end, this research employs a moderated mediation model as its core analytical framework. Within this model, critical thinking serves as the mediator variable to elucidate the internal pathway of the intervention's effect, AI dependency level functions as the moderator variable specifying the contextual conditions of this pathway, and creative thinking performance in instructional design constitutes the dependent variable for assessing the ultimate intervention outcome. The study is based on the theory of creativity investment (Sternberg & Lubart, 1991) and the theory of metacognition, and it is consistent with the principle of scaffolded learning in education (M. Chen & Tang, 2023).

Conceptual Framework and Operational Definitions: Creative instructional design is defined in this study as a professional practice process and its output in which teachers—by integrating subject matter knowledge, pedagogical knowledge, and creative thinking—develop comprehensive instructional plans aimed at achieving specific teaching objectives. These plans are characterized by both novelty (i.e., non-obvious, unique solutions) and educational utility (i.e., alignment with learner characteristics, practicality, and effectiveness in facilitating learning). It differs from mere "creative thinking" (a personal cognitive trait) and transcends "creative instructional design" that solely emphasizes novelty of outcomes. Instead, it underscores an integrated practical and design capability that unifies inventive conceptualization with instructional feasibility and effectiveness in authentic educational contexts. In this study, the term AI pedagogical agent refers specifically to a conversational intelligent agent powered by artificial intelligence and assigned the role of a "teaching partner". Its core function is to provide personalized support embedded with metacognitive scaffolding for pre-service teachers' instructional design tasks through structured interactions. It differs from general-purpose information retrieval chatbots or tutoring systems that merely offer standardized feedback. In essence, it is a strategic instrument purpose-built to implement a specific pedagogical intervention—namely, metacognitive scaffolding. Metacognitive scaffolding (MAS) originates from seminal theory of metacognition (often termed "cognition about cognition," which encompasses metacognitive knowledge and metacognitive monitoring). This study conceptualizes metacognitive scaffolding as an external support system embedded within the AI pedagogical agent's dialog, designed to guide cognitive processes. This system consists of structured prompts, guiding questions, and feedback frameworks (detailed in Appendix A) designed to assist learners in systematically reflecting on and managing their own cognitive processes—planning, monitoring, evaluating, and regulating—while completing instructional design tasks. Consequently, while grounded in the core tenets of the classical theory, the conceptualization of "metacognitive scaffolding" in this study has a clearly defined scope: it refers specifically to an external, structured support system targeting the domain-specific cognitive processes of instructional design. This delineates it from learners' implicit, general metacognitive abilities or broader conceptions of self-regulated learning.

2. Literature Review

2.1. Creativity and Teacher Education

Creativity is necessary for innovation and entrepreneurship (Lee, 2023; Yao et al., 2023). It is particularly important for those who will become teachers, because their ability to create new curricula and answer children's imaginations directly affects their success in the classroom in the future (Yildirim & Yilmaz, 2023). Early childhood is an important time for developing individuality that lasts into the future (Pan et al., 2022). Teachers' beliefs about creativity also affect educational practices during childhood (Agnoli et al., 2023). But traditional teacher education systems have long favored memorization and standard tests, often ignoring the development of students' creativity (Su et al., 2024). As the world changes and technology improves, we need more people with creative minds. Creative thinking is now more important than ever before, and it has become an absolute necessity in today's professions. Although the value of creativity is recognized, little is known about how students view the connection between new technologies, such as AI, and creativity (Marrone et al., 2022).

2.2. AI Pedagogical Agents in Education

AI's appearance brings big changes in the education area. Among them, AI pedagogical agents have attracted considerable attention due to their capacity to offer individualized advice and immediate feedback, especially within the realm of teacher education (Baglivo et al., 2023). They frequently employ LLMs to mimic the conduct of human mentors and give flexible learning courses and considerable educational aid (D. Chen et al., 2024). AI pedagogical agents that give personal guidance show the ideas behind personalized learning and connectivism, improving students' engagement (Alier et al., 2025). Hence, the mentorship philosophy seems to simply be about passing on knowledge, but rather also about aiding learners in coming up with their own ideas and solutions (Boye & Agyei, 2023). Nevertheless, presently, the teaching robots or agents of different AI tools focus chiefly on packaging the acquired knowledge and discipline training. Specifically, the current majority concentration is on science and math subjects. Such schooling ignores the development of complex cognitive thinking. Therefore, though we admit that technological progress has happened, it is important to note that the application of AI pedagogical agents in the development of creative instructional design is still in its infancy.

2.3. AI Dependency: A Potential Moderating Variable

Currently, education in particular dominantly puts forward AI dependence as a central governing agent requiring thorough investigation. Adopting AI brings not only opportunity but also downsides, a more concerning one being unparalleled reliance on technology. According to Creely and Blannin (2025), the overuse of AI in education may dictate students' learning experiences in such a way that their creativity and critical thinking are impeded. AI education must not just teach students how to collect answers from the internet but how to form a thinking process instead. This way, creativity will not be impaired by easy access to AI-generated answers. In fact, data show that AI technologies have differentiated outcomes. Consequently, the authors observe that multiple students frequently refer to generative AI for academic support; this has certainly added to concerns about AI's incorrect incorporation and the peril of extreme dependence. The major problem lies in finding out the basic essence of this coping strategy. The purpose here is not only to reduce superficial dependence, as represented by the acceptance of AI-generated content, but also to achieve direct conscious understanding of AI pedagogy. Students need to be led to view AI only as a mental tool that enables and amplifies human intellect, as opposed to the alternative, which is replacing it entirely.

2.4. The Theoretical and Practical Foundations of Metacognitive Awareness

The pedagogical principles of metacognitive awareness are the basis of improving learning quality with stronger evidence. According to M. Chen and Tang (2023), this awareness can be increase by granting learners help to describe and reinforce their own reasoning, mainly through given prompts with guiding questions and templates for reflection. For instance, a research teacher can encourage students through a sequence of questions, first setting the objective of the design task (planning), then prompting students to critically assess their plan (meta-evaluation) in order to create a new plan, and finally to examine alternatives (adjustment) that could have been implemented. However, traditional classrooms have barriers that prevent teachers from providing one-on-one, in-depth guidance throughout entire lessons. In turn, Xie et al. (2023) recommend that AI-fueled pedagogical helpers can successfully overcome this existing incapacity by promoting knowledge sharing and supporting students individually, ultimately contributing to a more integrated exchange between teachers and learners. Notwithstanding this capacity, the authors have noted an enduring skill gap among students; it is likely they can cope with basic tasks, but on the contrary, they may not have either the in-depth critical thinking or methods for the independent rapid improvement of their skills that robust metacognitive awareness training focuses on. Therefore, the dual action of integrating metacognitive awareness scaffolding into educational agents is advocated as a means to achieve these ends. Jiang et al. (2023) demonstrate this in their research, and say that such agents can be practically employed in structured metacognitive awareness training contexts, focusing on attention guiding and cognitive bias mitigation, along with evaluation and feedback-driven personalized learning. The intensity of the learning process is the main objective here, but parallel with it, the chances of out-of-control decisions when using AI technologies should be kept as low as possible.

2.5. Research Gaps and Questions

AI technology developing quickly brings great changes to education, especially teacher education. AI pedagogical agents hold huge potential. But most of the present AI pedagogical agent systems mainly concentrate on knowledge transmission and rudimentary skill training (Marrone et al., 2022), and there is little attention paid to fostering higher-order skills such as creative instructional design. Creative instructional design is one of the core competencies that pre-service preschool teachers should possess. Notably, none of the existing systems have integrated metacognitive awareness, which is a key support mechanism, even though metacognitive awareness is considered a fundamental foundation for the development of creative thinking (Jiang et al., 2023). In view of the above research gaps, this paper designs an AI pedagogical agent system integrating metacognitive awareness to explore the influence of metacognitive awareness on the creative thinking of pre-service preschool teachers. The system creates individualized metacognitive awareness suggestions and directs creative activities, helping pre-service preschool teachers' creative thinking procedures during course planning.

This research is grounded in robust theoretical frameworks. The creativity investment theory (Sternberg & Lubart, 1991) emphasizes that creative achievements result from the synergy of various cognitive resources. Metacognitive theory provides the framework for understanding the self-regulation processes involved in creative thinking. Based on these theories, we propose the following research questions and hypotheses:

RQ1. *Can an AI pedagogical agent based on MAS improve the performance of pre-service preschool teachers in creative instructional design?*

H1. *Compared with the control group using the conventional AI agent, the experimental group using the MAS-based AI agent will demonstrate significantly higher post-test scores on creative instructional design.*

RQ2. *How does the MAS-based AI pedagogical agent affect various aspects of creative thinking?*

H2. *MAS has different impacts on different aspects of creativity.*

RQ3. *How does critical thinking fit into the relationship between MAS and creative thinking?*

H3. *Critical thinking mediates the relationship between MAS and creative thinking. To be specific, MAS improves critical thinking, which then fosters creative thinking.*

RQ4. *How does AI dependency influence the function of MAS in creative thought?*

H4. *AI dependency will have a moderating effect on the relationship between MAS and students' creative thinking.*

3. Materials and Methods

3.1. Participants

The study participants came from the second year of a provincial level pre-service preschool teacher training college in China. A total of 120 participants were included in this study and were divided into a control group ($n = 60$) and an experimental group ($n = 60$). Regarding age, an independent samples t-test indicated no significant difference between the two groups. The control group (M = 19.07, SD = 0.31) and the experimental group (M = 19.05, SD = 0.29) were similar in age: $t(118) = 0.305$, $p = 0.761$, Cohen's d = 0.056. In terms of gender distribution, a chi-square test also showed no significant difference between the groups—$\chi^2(1) = 0.196$, $p = 0.658$, Cramer's V = 0.040—indicating comparable gender composition (control group: 20% male, 80% female; experimental group: 23.3% male, 76.7% female). To assess the level of AI experience, a self-developed AI Experience Scale (6 items, 7-point Likert scale) was used (Appendix C.4), which demonstrated good internal consistency (Cronbach's $\alpha = 0.907$). An independent samples t-test revealed no significant difference in AI experience between the control group (M = 5.35, SD = 0.92) and the experimental group (M = 5.24, SD = 0.72): $t(118) = 0.698$, $p = 0.487$, Cohen's d = 0.127. In summary, the two groups were comparable in terms of demographic variables and AI experience, satisfying the requirement for baseline equivalence in experimental grouping (see Table 1).

Table 1. Demographic and baseline characteristics of participants.

Variable	Group	*n*	Mean (SD) or %	Statistic	*p*	Effect Size
Age	Control	60	19.07 (0.31)	$t = 0.305$	0.761	d = 0.056
	Experimental	60	19.05 (0.29)			
Gender, % male	Control	60	20.0	$\chi^2 = 0.196$	0.658	Cramer's V = 0.04
	Experimental	60	23.3			

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Table 1. *Cont.*

Variable	Group	*n*	Mean (SD) or %	Statistic	*p*	Effect Size
AI experience level	Control	60	5.35 (0.92)	$t = 0.698$	0.487	d = 0.127
	Experimental	60	5.24 (0.72)			

3.2. Research Design

This study primarily employed a quasi-experimental design. Students were assigned to either the experimental group (using an AI pedagogical agent integrated with metacognitive strategies) or the control group (using a conventional AI pedagogical agent), with 60 participants in each group. The intervention was conducted over an eight-week period. This 8-week study was performed in Fall 2025 with pre-test, intervention, and post-test phases. All courses were taught by one teacher so that all students would have the same content and feedback. The instructor kept the SOP (Standard Operating Procedure) the same for both groups, as shown in Table 2, to reduce possible bias.

Table 2. Experimental procedure and design phases.

Phase	Time	Experimental Group ($n = 60$)	Control Group ($n = 60$)	Common Activities and Measurement Tools
Pre-Test	Week 1	60 min instructional design writing course	60 min instructional design writing course	All participants completed: • Demographic questionnaire • Metacognitive awareness scale • Critical thinking scale • AI dependency level scale • Creative thinking scale
Intervention	Week 2	60 min MAS pedagogical agent tool introduction and practice	60 min ordinary pedagogical agent tool introduction and practice	All participants completed platform exercises
	Weeks 3–7	Used MAS pedagogical agent tool to complete four-stage main instructional design tasks	Used ordinary pedagogical agent tool to complete four-stage main instructional design tasks	All participants: • Completed innovation task (designing "The Mysterious Clock") via AI pedagogical agent platform • Attended 40 min weekly instructional design course
Post-Test and Data Collection	Week 8	90 min: Post-test and questionnaires	90 min: Post-test and questionnaires	All participants: • Post-test scales: MAS, critical thinking, AI dependency level, creative thinking, innovation task performance • Data: AI dialog logs collected and analyzed
In-Depth Interviews	Week 8	Selected participants underwent 25–35 min semi-structured interviews		Interview content: • Experience during usage process • Analysis of usage effectiveness

In the first week, all participants completed demographic questionnaires, metacognitive awareness scales, critical thinking scales, AI dependence level scales, creative thinking scales, and baseline measures of innovative task performance.

The intervention period ran from week 2 to week 7. In week 8, the researchers measured both groups' responses to the metacognitive awareness scales, the critical thinking scales, the AI dependence level scales, and the creative thinking scales, as well as their creative instructional design task performance. AI dialog logs were gathered and frequency analysis was carried out. Some participants from the experimental group took part in semi-structured interviews (25–35 min each) to go into more depth about their learning process and results.

Before the study began, all participants signed a detailed informed consent form that explained the research goals, methods, possible dangers, and confidentiality rules. Participants were told clearly that they could leave the study at any time with no consequences so that it would be voluntary.

3.3. Teaching Materials

Both groups performed a new kind of teaching plan called "Making the Mysterious Clock" with the help of an AI pedagogical agent. All students took part in regular 90 min instructional design classes every week. The main difference between the experimental and control groups was that the control group had a normal AI pedagogical agent which gave them task-related Q&As based on knowledge, technical operation guidance, and standardized feedback, but no MAS prompts. The conventional AI agent provided direct answers and procedural guidance without prompting reflection. An example dialog is provided in Appendix A. The experimental group, however, employed a self-made MAS virtual platform and received special instruction on the platform's instructional design functions. The experimental group was told to use the platform in their task process to brainstorm, outline, optimize structure, and revise. The AI pedagogical agent used by the experimental group contained MAS. Each question asked of the experimental group was phrased as "what do you think about your design decision?" instead of giving the answer directly. The framework arranged the MAS procedures for the teachers in designing the "Mysterious Clock" activity. To guide teachers through continuous self-questioning, monitoring, and adjustment of their objectives, contents, methods, and procedures, the system aimed to improve their awareness, logic, and effectiveness in instructional design.

The metacognitive AI agent is not a generic conversational tool but is systematically designed to function as a metacognitive mentor for preschool teacher education in instructional design. Its core purpose is to guide student teachers, through Socratic questioning, to develop awareness, monitoring, and regulation of their own instructional design thinking, rather than delivering direct knowledge or solutions. This defined role ensures that every interaction is underpinned by a metacognitive orientation. Questioning Logic (Decision Rules): The agent's questioning follows a scaffolded progression from descriptive/factual to analytical/evaluative. It first prompts students to describe their initial ideas, then employs a series of "why," "how," and "what if" questions to guide them in-depth to analyze the theoretical rationale, child development characteristics, and pedagogical principles underlying their design decisions. Prompting Frequency and Triggering Mechanism: Prompts are not delivered at fixed intervals but are dynamically triggered based on key content milestones and the learner's cognitive state. For instance, when a student submits a preliminary objective, or when expressions such as "I'm not sure how to proceed" or "Is this design correct?" appear in the dialog log, the agent activates a series of metacognitive guiding questions, as in the following example dialog: Student Input: "I want to teach kindergarteners (5–6 years old) to recognize the clock, focusing on reading whole-hour times." Metacognitive Guidance: "That is a meaningful starting point. Let's reflect further: What is your rationale for setting 'reading whole-hour times' as the learning objective, based on the cognitive characteristics of children aged 5–6? In your view, what might be the main challenges and points of interest for children of this age in understanding the concept of time?" More details are available in Appendix A.

3.4. Measuring Tools

Metacognitive Awareness Scale(MAS): This scale was adapted from the Self-Report Scale created by O'Neil and Abedi (1996), with Cronbach's α of 0.876. It aims to evaluate one's own metacognitive awareness abilities via self-assessment. The scale measures how students usually perform when dealing with difficult problems, using a Likert scale (1 = never, 5 = always).

AI Dependency Scale (AIDT): The AIDT was created according to Hou et al. (2025), intended to evaluate students' usual interaction methods with generative AI when solving difficult problems. The scale has four behavioral dimensions: reflective use, which means actively adjusting and deeply integrating AI output; cautious use, highlighting the recognition and prevention of possible AI mistakes; reckless use, showing passive dependence on AI with little cognitive involvement; and collaborative use, centered around AI usage among peers. These are evaluted using a Likert scale (1 = never, 5 = always). Higher scores indicate more frequent engage in such dependency behavior. The total scale has good reliability, with Cronbach's alpha of 0.835. Validity-wise, the four-dimensional structure of the scale was confirmed by EFA and CFA, which indicates that it can be applied to different types of complex problem-solving situations. It only looks at what people do when they interact with things; it does not check how well they do their work.

Critical Thinking Disposition Inventory (CCTDI): This scale is based on the California Critical Thinking Disposition Inventory (Facione & Facione, 1992) and follows classical critical thinking theory. It was made to assess people's readiness to and habits of applying critical thinking in their everyday lives, utilizing a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Here, critical thinking is seen both as a mental skill and as a lasting intellectual attitude, marked by a readiness to think and consider things. The overall scale has good reliability, with Cronbach's α of 0.865.

Creative Thinking Scale (CTS): The scale that was used for this study was an adapted version of the Williams Creativity Scale used to measure individual creativity. Overall scale reliability is good, with Cronbach's α of 0.895. The tool consists of five main aspects: adventurousness (willingness to take on uncertainty and explore the unknown), curiosity (sustained intrinsic desire to learn about how things work), imagination (ability to form and visualize non-realistic situations), fluency (capacity to produce a substantial amount of ideas, perspectives, or solutions over a certain period), and challenging (tendency to recognize and tackle difficult issues). Participants are asked to rate a set of statements about their behavior, attitude, and preference on a 5-point Likert scale (1 = "Not at all" to 5 = "Completely"). A higher score means a higher level of creativity in that area.

Innovative Teaching Design Performance: The students had to hand in their final "Clock recognition" course design plan, which was graded out of 100 points by two educators who did not know the groups. Scoring was based on "novelty" and "teaching practicality." To ensure scoring consistency, inter-rater reliability was assessed using the Intraclass Correlation Coefficient (ICC > 0.8).

4. Data Analysis

4.1. Quantitative Data Analysis

For quantitative data, an independent samples *t*-test was conducted to compare the creativity levels and task performance of the experimental and control groups in the pretest, checking for significant improvement. MANOVA was used to test if there was a significant difference between the two groups' overall creativity level, which included five dimensions. To examine the mediating role of critical thinking and the moderating role of AI dependency on the link between MAS and creative instructional design, we used the PROCESS macro (Model 14) created by Hayes (2022), deploying a bias-corrected percentile bootstrap with 5000 resamples; 95% CI not including zero indicates significant mediation.

4.2. Qualitative Data Analysis

To gain an in-depth, learner-centered understanding of the experience, effectiveness, and mechanisms of the MAS intervention, semi-structured interviews were conducted with participants from the experimental group after the study.

Sampling and Implementation: A maximum-variation sampling strategy was employed. Based on participants' performance in the post-test creative instructional design task (top 25%, middle 50%, bottom 25%), 15 representative students were selected for interviews. Each interview lasted 25–35 min. The interview protocol was designed around two core dimensions (see Appendix C.5 for details): the experiential dimensions of the process (including planning, monitoring, regulation, and evaluation) and the perceived outcomes of use (including knowledge mastery, motivational confidence, and transferability of application).

Data Analysis Procedure: Interview recordings were transcribed verbatim. The transcripts were then systematically analyzed using thematic analysis, assisted by the qualitative data analysis software NVivo14. The procedure consisted of the following stages: ① Familiarization and Initial Coding: Two researchers first immersed themselves in the data by repeatedly reading the transcripts. Guided by the interview protocol, they independently performed open coding on statements related to MAS, creative thinking, AI dependency, etc. (For example, the statement "It helped me deconstruct vague course objectives into specific teaching steps" was coded as "1.1 Deconstruct Core Difficulties".) ② Theme Development: Through discussion, the researchers collated and grouped the numerous initial codes to form more abstract secondary and tertiary themes. (For instance, initial codes such as "1.1 Deconstruct Core Difficulties," "1.4 Enhance Time Allocation," and "1.11 Plan Updates Timely" were clustered under the secondary theme "1. Adjust Plan Values," ultimately leading to the tertiary theme "Enhancing MAS capabilities during the design process, and strengthening creative instructional design skills throughout the process." The complete coding scheme is presented in Table A7, Appendix D) ③ Reviewing and Refining Themes: The research team collectively reviewed and discussed the thematic structure, naming, and representativeness. Discrepancies were negotiated until consensus was reached. To assess coding consistency, an inter-coder reliability test was conducted on a randomly selected 20% of the transcripts using Cohen's Kappa coefficient. The Kappa value for all major themes was 0.86. ④ Reporting: The finalized themes were integrated with vivid and representative interview excerpts to construct the qualitative findings.

5. Results

5.1. Innovative Teaching Design Task Performance

To examine the effect of the instructional intervention on teachers' instructional design task performance, an independent samples t-test was conducted between the experimental group ($n = 60$) and the control group ($n = 60$). As shown in Table 3, the experimental group scored higher (M = 85.75, SD = 1.39) on the instructional design task performance measure than the control group (M = 85.08, SD = 0.81). The difference between the two groups was statistically significant—$t(95.02) = 3.22$, $p = 0.002$—with the experimental group outperforming the control group. In summary, the instructional intervention significantly enhanced teachers' performance in instructional design tasks, and the effect was practically meaningful.

Table 3. Independent samples t-test results.

Group	N	M	SD	t	df	p
Experimental Group	60	85.75	1.39	3.22	95.02	0.002
Control Group	60	85.08	0.81			

5.2. Impact of Metacognitive Scaffolding Intervention on Different Aspects of Creative Thinking

To examine the overall predictive effect of metacognition on the various dimensions of creative thinking, a multivariate analysis of variance (MANOVA) was conducted.

The results showed that metacognition had a significant overall predictive effect on the five dimensions of creative thinking (curiosity, imagination, challenge, risk-taking, and fluency)—Wilks' Λ = 0.689, F = 4.873, $p < 0.001$, partial $\eta^2 = 0.311$ (Table 4).

Table 4. Results of multivariate analysis of variance (MANOVA) test.

Effect	Test Statistic	Value	F	Hypothesis df	Error df	p	ηp^2
MAS	Wilks' Λ	0.689	4.873	5	54	<0.001	0.311

The result revealed that the specific predictive effects of metacognition on each dimension of creative thinking varied. Ranked by effect size (partial η^2) from largest to smallest, the results were as follows: metacognition had the most prominent predictive effect on curiosity—F (1, 118) = 63.668, $p < 0.001$, partial $\eta^2 = 0.350$; its predictive effect on risk-taking was comparable—F (1, 118) = 63.559, $p < 0.001$, partial $\eta^2 = 0.350$; this was followed by its effect on challenge—F (1, 118) = 54.134, $p < 0.001$, partial $\eta^2 = 0.314$; then its effect on fluency—F (1, 118) = 48.471, $p < 0.001$, partial $\eta^2 = 0.291$; and its effect on imagination had the smallest effect size among the dimensions, though still highly significant—F (1, 118) = 45.399, $p < 0.001$, partial $\eta^2 = 0.278$. All effect sizes fell within the range indicating a large effect, demonstrating that metacognition possesses substantial and important predictive power for each dimension of creative thinking (Table 5).

Table 5. Multivariate test results of MAS intervention on dimensions of creative thinking.

Dependent Variable	p-Value	Partial η^2	Statistical Conclusion
Curiosity	<0.001	0.350	significant
Risk-taking	<0.001	0.350	significant
Challenge	<0.001	0.314	significant
Fluency	<0.001	0.291	Significant
Imagination	<0.001	0.278	Significant

5.3. *The Impact of Critical Thinking and AI Reliance*

To examine whether AI reliance level moderated the mediating role of critical thinking between metacognition and creative thinking, Hayes' (2022) PROCESS macro (Model 14) was employed with 5000 bootstrap resamples and a 95% confidence level. The first-stage regression analysis (see Table 6) showed that metacognition had a significant positive predictive effect on the mediator, critical thinking ($\beta = 0.793$, $p < 0.001$). The second-stage regression analysis indicated that the interaction between critical thinking and AI reliance level significantly predicted creative thinking ($\beta = 0.129$, $p = 0.008$), suggesting that AI reliance level significantly moderates the relationship between critical thinking and creative thinking. The direct effect of metacognition on creative thinking approached, but did not reach, conventional levels of statistical significance ($\beta = 0.227$, $p = 0.088$), whereas the indirect effect was significant. In summary, critical thinking mediates the relationship between metacognition and creative thinking, and this mediating effect is positively moderated by AI reliance level.

Table 6. Results of moderated mediation analysis.

Variable	β	SE	t	p	95% CI	Conclusion
Stage 1: Dependent Variable = Critical Thinking						
Constant	12.190	2.374	5.134	<0.001	[7.488, 16.892]	Significant
MAS	0.793	0.055	14.467	<0.001	[0.685, 0.902]	Significant

Table 6. *Cont.*

Variable	β	SE	t	p	95% CI	Conclusion
Stage 2: Dependent Variable = Creative Thinking						
Constant	8.572	4.281	2.003	0.048	[0.093, 17.051]	Significant
X (Direct Effect)	0.227	0.132	1.722	0.088	[−0.034, 0.488]	Not Statistically Significant
M	0.709	0.133	5.314	<0.001	[0.445, 0.974]	Significant
W	−5.775	2.343	−2.464	0.015	[−10.417, −1.133]	Significant
M × W (Interaction Term)	0.129	0.048	2.704	0.008	[0.035, 0.223]	Significant

Note: X = MAS, M = Critical Thinking, W = AI Reliance. Bootstrap sample size = 5000. A confidence interval that does not include zero indicates a significant effect.

5.4. Qualitative Analysis and Results

The interview results show that students had a better understanding of the subject, more enthusiasm and confidence, and better transferable skills after the intervention (see Appendix D). One student said, "After using the AI pedagogical agent, my planning habits for preparing lessons and writing lesson plans became more focused and actionable." The main change is obvious in moving from a fuzzy framework to specific breakdowns. "When I hit a cognitive block in my learning process, such as getting stuck on designing teaching activities, I would turn to the AI pedagogical agent for some help and guidance, which would clear up my confusion. I feel that this project has helped guide my learning and increase my confidence in teaching. It has helped me understand the teaching process and come up with a strategy that allows me to learn better, which will have a positive effect." They said that the AI agent made up for the gaps in their knowledge, lacking in diversity, by providing them with sample plans or superiority plans as alternatives. The offered support not only ignited their imaginative cogitation but also nurtured a cycle in which the designers kept asking more and more questions and outlined new versions of their design in the process, ultimately with more original and practical ideas. These observations are coherent with the mediation process, in which MAS boosting critical thinking further leads to creative breakthrough, as it unfolds in real learning situations.

Nonetheless, while there were some laudable results, students also had some big reservations. The prevalent concern was primarily related to the problem of over-reliance on artificial intelligence. One participant made this concern evident to an even larger extent, saying, "I am frightened about my getting more and more adjustment to AI, which may result in the lessened mode of advanced design work controlling."

6. Discussion

Employing a quasi-experimental design, this research investigated how metacognitive awareness strategy scaffolding, when integrated into an AI pedagogical agent, influences creative thinking development among pre-service early childhood education students. In this study, we deduced an outline that elaborates on the four major lessons that may be learned. First, concerning the application of an intelligent agent's support during a novel instructional design task, the intervention group achieved substantially greater results in comparison with the conventional AI tutoring system (supporting Hypothesis 1). Second, metacognition exerts a significant influence across all dimensions of creativity, albeit to varying degrees across different dimensions (providing partial support for H2). Specifically, its impact on curiosity, risk-taking, and challenge-seeking is somewhat more pronounced than on fluency and imagination. Third, critical thinking served as an intermediary affecting the relationship between MAS exposure and creativity. Regarding learning, the present findings specify this factor as a specialized route: MAS fosters critical thinking, which contributes to the emergence of creativity (realizing the major point for H3). Fourth, AI dependency was shown to be a moderating factor in the finding (realizing the major point of H4). Taken together, these findings construct an empirical framework that can be utilized for designing AI-enhanced instructional environments capable of fostering higher-order cognitive abilities.

6.1. MAS as a Meta-Scaffold That Gives Boost to Creative Thinking in Human Beings

In contrast to the experimental instruction developed under the control conditions with a traditional AI instructor, MAS chaining was used as a complementary teaching approach that showed an impressive progression in creative instruction design assessment—an outcome supporting H1. This indicates the challenge of Machine Learning in education, as knowledge transfer is now understood as a weakness of artificial intelligence, while higher cognitive skills, such as designing or debating using knowledge, have become its goal (Marrone et al., 2022). Furthermore, it strengthens the idea that designing a computer-based system with visual instructions may be more advanced than simply making the computer a common tool for students. On the other hand, students should focus on the benefits of using AI not only in basic activities but also in encouraging logic and problem-solving development.

For instance, interview results indicated that students moved from a "confused plan" to a "specific work breakdown" as they proceeded in their learning. Furthermore, reproducible terms such as "children," "ability," and "observation" generally resulted in a major shift in the learners' cognitive approaches. Particularly, MAS supports this by issuing constraints and framing the mental processes of the students, which combines the internal thoughts and translates into the external actions of the designers of pedagogical plans. This aligns with the foundational theory of MAS, that contemplation of one's own thinking is critical for the result of cognitive efficiency. Moreover, the fact that creative thinking consists of such a real-time activity, whose regulation and management are provided by the MAS mechanisms, was confirmed in a recent study (Jiang et al., 2023). This framework, grounded in the theory of learning through artificial intelligence-enhanced learning, is used within an entirely new domain of AI-based education.

The results of the multivariate analysis not only deepen our understanding of the predictive capacity of the MAS mechanism, but also reveal significant differences in the strength of its effects across various domains of creativity (providing partial support for H2). MAS demonstrates a pronounced role in predicting curiosity, challenge-seeking, and risk-taking. Its core value lies in stimulating institutionalized motivational planning and fostering learners' self-regulatory development—specifically, igniting the curiosity to explore novel concepts, the courage to confront complex tasks, and the adventurous spirit to overcome doubts. However, its association with ideational fluency is relatively weak. This suggests that while MAS does not directly facilitate the generation of a large quantity of ideas, it significantly enhances the depth of exploration and the quality of the idea formation process.

This differentiation carries important theoretical implications: MAS should not be regarded as a "universal enhancer" capable of improving all dimensions of creativity uniformly. Instead, it functions more as a "cognitive navigation system"—it sets exploratory goals, strengthens intellectual curiosity, and supports responses to unknown challenges. While it does not directly supply the "inspiration prototypes" required for ideational fluency, it charts a cognitive blueprint for the realization of original thinking. This insight holds significance for future evaluations of educational technology: when assessing the impact of teaching agents and other educational technologies on creativity, adopting a multidimensional analytical framework is essential. Such an approach helps prevent the masking of differential responses across specific creative dimensions due to an overemphasis on "overall effects" (Marrone et al., 2022; Jiang et al., 2023).

6.2. The Mediating Role of Critical Thinking and the Moderating Effect of AI Dependence

This study identifies the critical thinking pathway, in its mediating role, to be the sole causative factor that establishes the relation between AI pedagogical agent management and creative thinking (supporting H3). In short, MAS does not inspire creativity, per se. Thus, it transcends usual academic skills and provides students with tools to help them be creative thinkers. Through the identified pathway "MAS → Critical Thinking → Creative Thinking," a clear cognitive development line can be drawn. Although the direct effect was not statistically significant (p = 0.088), the positive coefficient and its relatively low p-value might suggest a potential trend that warrants further investigation in future studies with larger samples.

The positive moderating mechanism of AI reliance may be one of the following: ① Augmentation Assistance: AI tools may provide richer information and analytical support, enabling deeper and more comprehensive critical thinking, which in turn facilitates the stimulation of creative thinking. ② Cognitive Offloading: AI reliance may help individuals handle lower-level cognitive tasks, freeing up cognitive resources for higher-level creative thinking activities. ③ Interactive Inspiration: During interactions with AI, feedback and suggestions from AI may inspire new perspectives, thereby enhancing the translation of critical thinking into creative thinking. AI reliance positively moderates the pathway from critical thinking to creative thinking, suggesting an interaction effect between AI reliance and critical thinking: when critical thinking is high, AI reliance may promote creative thinking; when critical thinking is low, AI reliance may inhibit creative thinking (due to the negative main effect). Thus, the role of AI reliance is complex: it may directly suppress creative thinking, yet indirectly promote it by enhancing the effect of critical thinking. The overall impact depends on the level of critical thinking. Qualitative data also surfaced a critical caveat: students voiced apprehension about becoming overly dependent on AI, fearing a loss of control over their own design processes. This suggests that while AI dependency may statistically enhance the creativity pathway, it also introduces a psychological and pedagogical risk of diminishing learners' sense of agency and metacognitive autonomy. Thus, the moderating role of AI dependency should be interpreted with caution—it may facilitate creativity in the short term but could potentially inhibit deeper, self-sustained creative development if not accompanied by deliberate pedagogical strategies to foster cognitive ownership.

7. Conclusions and Implications

7.1. Conclusions

In summary, this study provides empirical evidence that an AI pedagogical agent integrated with metacognitive scaffolding (MAS) can enhance the creative instructional design performance of pre-service preschool teachers. The results support a moderated mediation mechanism, whereby the positive impact of MAS on creativity appears to be primarily channeled through the enhancement of critical thinking. Furthermore, this mediating pathway is positively moderated by learners' AI dependency level. Notably, the findings highlight the potential "double-edged sword" effect of AI reliance in fostering creativity. Although the findings are insightful, they should be regarded as a preliminary framework and empirical starting point rather than a universal or definitive solution. Their validity and generalizability warrant further examination across different disciplines, cultural contexts, and extended time spans.

7.2. Limitations and Future Research Directions

Though it has made some contributions, the study still has its limitations. Participants were pre-service early childhood education majors from one institution, which limits the generalizability to other disciplines and cultures. The eight-week intervention documents the short-term effects; whether the MAS habits and creative thinking gains are durable is unknown. Additionally, the limited sample size of this study necessitates that the robustness and generalizability of the findings be further validated in future research. Critical thinking was mainly measured by self-report dispositions; more work is needed to include performance-based behavioral measures to assess skills. Lastly, the model focused on individual cognitive and motivational factors and ignored peer dynamics and larger social context.

Future research should therefore pursue several directions with the following aims: to conduct longitudinal studies to follow the long-term course and transfer of MAS-based interventions; to test the generalizability of the model to other subjects (such as STEM and humanities teachers' education) and different cultural groups; to create integrated interventions combining MAS with affective and collaborative scaffolding; and to use fine-grained approaches such as eye-tracking and neuroimaging to uncover the immediate cognitive–neural processes by which MAS impacts creativity.

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Abbreviations

The following abbreviations are used in this manuscript:

MAS	Metacognitive Awareness Scaffolding
AIDT	AI Dependency Test
CCTDI	California Critical Thinking Disposition Inventory
CTS	Creative Thinking Scale

Appendix A

Table A1. MAS guidance process design framework—taking "The Magical Clock" as an example.

Stage	Core Task	MAS Guiding Questions	Design Points and Reflection Keys
Phase 1: Goal-Setting	Clarify teaching direction and rationale	1.1 Identifying the Starting Point for Thinking: When choosing the "clock" theme, what are the primary developmental needs of preschoolers in the senior class you must first identify? Are these needs appropriate for the cognitive characteristics of 5–6-year-old children? What is the basis for your judgment? 1.2 Reflecting on Goal-Setting Strategy: From which dimension(s) do you plan to set the objectives (knowledge/skill/affect)? Why choose these dimensions over others? Please describe your decision-making process.	• Link mathematical cognition (concept of time), living habits (regular routine), and social development (sense of time). • Based on Piaget's assumptions about cognitive development, judge that young children are at the end of the preoperational stage, capable of understanding concrete symbols, but abstract thinking requires the aid of concrete objects. • Objectives should reflect the integrated development of cognitive recognition, motor skills, and affective attitudes, avoiding a singular focus on knowledge transfer.
Phase 2: Content Selection	Deconstruct knowledge and connect to experience	2.1 Concept Deconstruction Analysis: How would you break down "clock recognition" into sub-concepts (e.g., function of hands, units of time, etc.)? What is the logical sequence for ordering these concepts? Which ones might become cognitive difficulties? 2.2 Connecting to Children's Experience: How can the abstract concept of time be connected to children's daily life experiences? What cognitive conflicts do you anticipate children might have? How will you address them?	• Deconstruction sequence: Overall recognition of the clock → numbers (1–12) → hour hand (short, slow) → minute hand (long, fast) → o'clock time → half-past. • Difficulties: Understanding the hour hand's position at half-past; the abstract nature of the "minute" unit. • Connection strategy: Directly associate "what time" with "what to do" (e.g., time for snacks, 12 o'clock lunch). • Anticipated conflict: Why does the hour hand only move one notch when the minute hand goes all the way around? Resolve through embodied manipulation (role-playing the clock hands).
Phase 3: Method Design	Select strategies and prepare materials	3.1 Justifying Choice of Methods: Why do you consider the hands-on operation method/game-based method more suitable for this theme than the lecture method? If a child consistently fails to understand the half-past concept, what is your alternative strategy? 3.2 Reflecting on Material Design: When making a DIY clock teaching aid, how do you ensure the pointer rotation mechanism meets safety standards? What considerations factor into choosing a real analog clock vs. a digital clock for demonstration?	• Justification: Young children's thinking is characterized by concrete imagery; hands-on operation and games facilitate active exploration and experiential learning. • Alternative strategy: Return to life events (e.g., naptime is at 11:30); use visual aids (e.g., dividing the clock face into colored sections). • Safety standards: No sharp protrusions where pointers connect to the face; appropriate rotation tightness to prevent detachment or pinching. • Considerations: Analog clocks more intuitively demonstrate the 联动关系 (linking movement) of the hands; digital clocks connect to modern life experiences. Both can be used in combination.
Phase 4: Process Optimization	Plan timing and provide differentiated support	4.1 Time Allocation Decisions: What is the rationale for the time proportion allocated to each segment (Introduction/Exploration/Summary)? If the actual operation runs overtime, which segment would you prioritize compressing, and why? 4.2 Considering Differentiated Design: How would you design extension challenges for children who progress faster? What scaffolded support would you prepare for children struggling with number recognition?	• Time allocation: Exploration (approx. 50%) > Introduction (approx. 30%) > Summary (approx. 20%). The rationale is that children's independent exploration is the primary pathway for learning. • Compression priority: Prioritize compressing teacher-led introduction or summary segments to protect children's core exploration and operation time. • Extension challenges: Design tasks like creating "My Daily Schedule" or challenging them to understand "quarter-hour" concepts. • Scaffolded support: Provide clocks with dot markers for counting, number matching cards, or opportunities for one-on-one peer assistance.

Table A2. Dialog framework for control-group AI agent.

Step	User Input	AI Response
1	"I want to design a clock lesson."	"OK. Steps: (1) Show a clock; (2) Explain hands; (3) Teach whole hours."
2	"What should 5–6 year olds learn?"	"Goal: recognize clock face and whole hours. Teach in order: clock → hands → hours."
3	"How should I teach it?"	"Methods: (1) Hands-on; (2) Games; (3) Songs. Pick one."
4	"How to make a DIY clock?"	"Use cardboard and split pins. Keep hands smooth and pins secure."
5	"How long should it take?"	"35 min: intro (5), explain (10), activity (15), close (5)."
6	"I'll use a game."	"Good. Games work. Keep rules simple."

Note: The control agent gave direct answers, no reflection prompts.

Appendix B

Detailed CFA Results Software: SPSS27 estimator: Maximum Likelihood.

Table A3. Metacognitive awareness scale results (12-item scale). Cronbach's α = 0.876.

KMO and Bartlett's Test		
KMO Measure of Sampling Adequacy		0.889
Bartlett's Test of Sphericity	Approx. Chi-Square	477.676
	df	66
	Sig.	<0.001

Table A4. AI dependency tendency scale results (16-item scale). Cronbach's α = 0.835.

KMO and Bartlett's Test		
KMO Measure of Sampling Adequacy		0.790
Bartlett's Test of Sphericity	Approx. Chi-Square	566.257
	df	120
	Sig.	<0.001

Table A5. Creative thinking scale (CTS) results (15-item scale). Cronbach's α = 0.895.

KMO and Bartlett's Test		
KMO Measure of Sampling Adequacy		0.890
Bartlett's Test of Sphericity	Approx. Chi-Square	665.816
	df	105
	Sig.	<0.001

Table A6. Critical thinking scale (CCIST) results (13-item scale). Cronbach's α = 0.865.

KMO and Bartlett's Test		
KMO Measure of Sampling Adequacy		0.853
Bartlett's Test of Sphericity	Approx. Chi-Square	524.173
	df	78
	Sig.	<0.001

Appendix C

Appendix C.1

Adapted metacognitive awareness scale—12 items (focused on innovative teaching design).

1. Before starting my teaching design, I first consider the best strategies or methods.
2. I set specific goals and plans for my teaching design.
3. While designing, I can clearly recognize what content I have mastered and what I have not.
4. When I feel mentally overwhelmed, I can realize that I need to pause and reorganize my thoughts.
5. I regularly check my progress toward my goals and adjust my strategies as needed.
6. When solving design problems, I consciously consider whether the thinking methods I am using are effective.
7. I am good at selecting useful information while ignoring irrelevant distractions.
8. After answering questions or completing tasks, I review and assess my performance.
9. When I cannot understand a concept, I know where to look for answers or seek help.
10. I reflect on my learning or thinking processes and summarize which methods work best for me.
11. While designing, I continuously summarize key points in my mind to ensure I keep up with the pace.
12. I can anticipate the potential difficulties of teaching design tasks and allocate my time and energy accordingly.

Appendix C.2

Adapted critical thinking scale—13 items (focused on innovative teaching design).

1. I enjoy tackling teaching design problems that require deep thinking to solve.
2. Before making decisions, I try to consider various alternatives.
3. When an AI pedagogical agent attempts to persuade me with emotional appeals (such as empathy), I remain vigilant.
4. I am confident in my ability to design teaching strategies.
5. When I discover that my viewpoint is incorrect, I am willing to acknowledge and change it.
6. I believe that any viewpoint or belief should withstand questioning and scrutiny.
7. In discussions, I am adept at posing deep questions that provoke further thought.
8. I strive to identify biases, assumptions, and logical fallacies in the statements made by AI pedagogical agents.
9. When faced with a difficult problem, I try to examine it from multiple perspectives.
10. I believe that most complex problems do not have a single "correct answer."
11. During conversations with AI, I pay attention to distinguishing between factual statements and personal opinions.
12. Even when time is tight, I tend to take the time to think things through rather than act hastily.
13. I prefer to plan teaching design activities more thoroughly and strategically.

Appendix C.3

Adapted creative thinking scale—15 items (focused on innovative teaching design).

Curiosity

1. I often feel curious about the principles, reasons, or mechanisms behind teaching design and actively explore them.
2. I question existing methods during the design process and wonder, "Is there a better way?"
3. I can easily connect knowledge or concepts from different fields to generate new ideas during the design phase.

Imagination

4. I frequently engage in "brainstorming" in my mind, imagining various possible scenarios or stories.
5. I enjoy expressing abstract ideas in my designs through drawings, charts, or concrete representations.
6. I like participating in activities that require imagination (e.g., role-playing, improvisation).

Challenge

7. I am skilled at identifying subtle issues or details in others' teaching designs.
8. I feel excited rather than intimidated when facing challenges in innovative teaching design.
9. I enjoy the process of solving tricky problems in teaching.

Risk-taking

10. I am willing to try new approaches to solving design problems, even if they may fail.
11. I am not afraid to propose ideas that might be considered "silly" or "outlandish" during teaching design.
12. Even in the face of criticism or opposition, I am willing to stand by my innovative ideas.

Fluency

13. When I encounter a problem, many different solutions come to mind.
14. I enjoy trying out various teaching activities during the design process.
15. I often think of unusual uses for teaching methods.

Appendix C.4

Adapted ai experience scale—six items (focused on AI usage experience).

1. I can recognize the artificial intelligence technologies used in the applications and products I use.
2. I am proficient in using artificial intelligence applications or products to assist daily work.
3. I can use artificial intelligence applications or products to improve work efficiency.
4. After using them for a period of time, I can evaluate the capabilities and limitations of artificial intelligence applications or products.
5. I can select appropriate solutions from multiple intelligent modeling solutions.
6. I can choose the most suitable artificial intelligence application or product for a specific task.

Appendix C.5

Interview protocol:

Process Experience:

(Planning) Have you noticed any changes in your habit of planning before you start a learning activity after using the AI pedagogical agent? Can you explain more about it?

(Monitoring) During the learning process, if you have problems or lose concentration, can the AI pedagogical agent make you realize it earlier? How do you respond?

Can you give me an instance where the pedagogic agent led you to modify your first learning strategy or teaching method?

(Evaluation) Have you noticed any changes in how often or deeply you reflect on and summarize what you have learned after finishing a learning activity? In what way did the AI pedagogical agent help with this?

Usage Effectiveness:

(Knowledge Mastery): Have you felt that the AI pedagogical agent has helped you understand the subject matter better or the art of teaching?

(Motivation and Confidence): Does this project affect your motivation to study and your confidence in becoming a qualified teacher? Is it positive or negative?

(Transfer of Application): Have you used the metacognitive strategies that were developed with the help of the AI pedagogical agent in other courses or for self-study?

Appendix D

Table A7. Qualitative coding scheme.

Primary Coding	Secondary Coding	Tertiary Coding
1.1 Deconstruct Core Difficulties 1.2 Optimize System Countermeasures 1.3 Improve System Efficiency and Operability 1.4 Enhance Time Allocation 1.5 Improve Planning Robustness 1.6 Guide Analysis Balance 1.7 Design Conforms to Educational Principles 1.8 Plan Update Measures 1.9 Strategy Optimization from the Beginning 1.10 Improve Plan Implementation Rate 1.11 Plan Updates Timely 1.12 Plan Contingency Strategies in Advance 1.13 Supplement Interactive Sessions 1.14 Meet Needs 1.15 Propose Guidance Plan 1.16 Systematic Planning 1.17 Solve Main Problems 1.18 Update within a Set Timeframe 1.19 Planning Requirements 1.20 Improve System Efficiency 1.21 Facilitate Plan Implementation 1.22 Improve System Scientificity 1.23 Systematize Each Process 1.24 Provide Suggestions 1.25 Incorporate Designs Suitable for Young Children	1. Adjust Plan Values	Enhancing metacognitive awareness during the design process, and strengthening creative instructional design skills throughout the process.
2.1 Remind to focus on task theme 2.2 Provide timely reminders 2.3 Remind to concentrate attention 2.4 Monitor learning progress 2.5 Redirect attention back to focus 2.6 Proactively guide in problem-solving 2.7 Identify shortcomings/areas for improvement	2. Enhancing Monitoring	
3.1 Provide quick answers and clarification 3.2 Give explanatory feedback 3.3 Help adjust mindset and refocus on content 3.4 Guide adjustment of thinking approaches 3.5 Suggest specific revisions	3. Enhancing Monitoring	
4.1 Reflect on strategy effectiveness 4.2 Focus reflection on practical problems 4.3 Engage in reflection and revision 4.4 Conduct active review and summarization 4.5 Push structured reflection checklists 4.6 Promote more systematic reflection	4. Enhancing Reflection and Evaluation	

Table A7. *Cont.*

Primary Coding	Secondary Coding	Tertiary Coding
5.1 Deepen understanding of knowledge 5.2 Understand disciplinary knowledge 5.3 Grasp knowledge efficiently 5.4 Link theory with practice 5.5 Deconstruct complex knowledge 5.6 Make abstract knowledge more intuitive 5.7 Develop a profound understanding of knowledge 5.8 Promote depth of knowledge comprehension	5. Enhancing Knowledge Master	Improving innovative design effectiveness and promoting professional development
6.1 Increase confidence 6.2 Gain a sense of achievement 6.3 Strengthen sense of competence	6. Enhancing Motivation and Confidence	
7.1 Facilitate learning of other knowledge 7.2 Apply critical thinking to analyze other works	7. Promoting Transfer and Application	

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